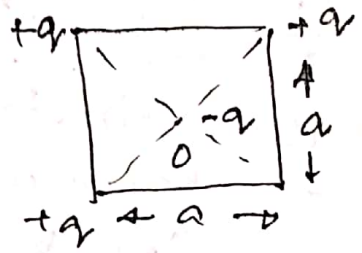
Problems on Coulomb's Law

Homework set-1.

(1) Two charged bodies (point charges) are kept at separation of 1m. What is minimum electrostatic force betn. them? $\left[\frac{e^2}{4\pi\epsilon_0}\right]$

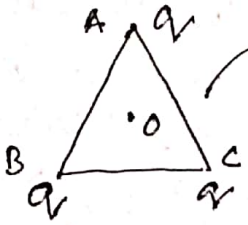
(2)



Find electric force on '-q' at O. $\left[\frac{q^2}{2\pi\epsilon_0 a^2}\right]$

(3) Five point charges each '+q' are placed on five vertices of a regular hexagon of side 'a' m. What is magnitude of force on a point charge of value '-q' placed at centre of hexagon.

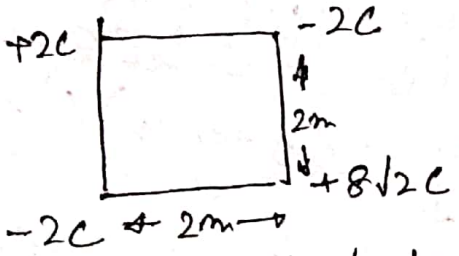
(4)



equilateral triangle. What will happen to charge q at A if a pt charge -q is placed at centre? [will it be at rest] If not for what value of charge at O, the charges at vertices will be at rest. [no moves to O, $-\frac{q}{\sqrt{3}}$]

(5) Four charges each '+q' are placed at four corners of a square. Find nature, position & magnitude of a 5th charge for which charges at vertices will be at rest. [-ve, centre, $\frac{q}{4}(1+2\sqrt{2})$]

(6)

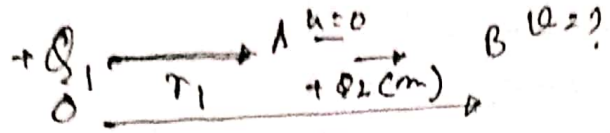


Find force on +2C [$K\sqrt{2}$, $K = \frac{1}{4\pi\epsilon_0}$]

(7) A charge 'Q' is to be divided on two objects. What should be the value of charges on the objects so that force betn. them is maximum? [$Q/2$, $Q/2$]

Scanned with CamScanner

(8)



on a smooth surface charge $+Q_1$ is at rest & charge $+Q_2$ (mass m) starts from rest at A ($OA = r_1$). What will be its velocity after reaching pt B ($OB = r_2$)

$$\left[\sqrt{\frac{2KQ_1Q_2}{m}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)}\right]$$ (Done in video class 1)

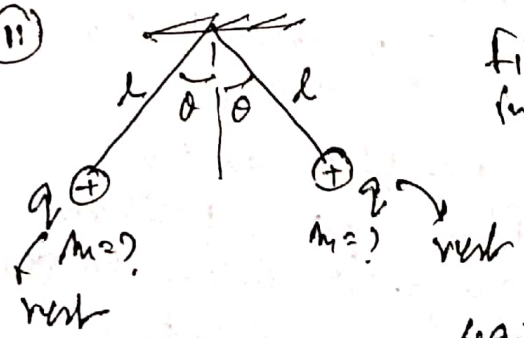
(9) A particle of charge 'q' is placed directly below at a distance 'r' from a stationary simple pendulum bob of mass 'm'. What should be the charge on the bob so that the string is loose?

$$\left[\frac{mgr^2}{Kq}\right] \quad K = \frac{1}{4\pi\epsilon_0}.$$

(10) Two pt charges each of charge (+q) & mass 'm' are suspended by nonconducting threads of length 'l' each. If angle betn. threads is very small then show that at equilibrium distance betn. two charges is $x = \left(\frac{2q^2 l}{Kmg}\right)^{1/3}$

$$\left(K = \frac{1}{4\pi\epsilon_0}\right)$$

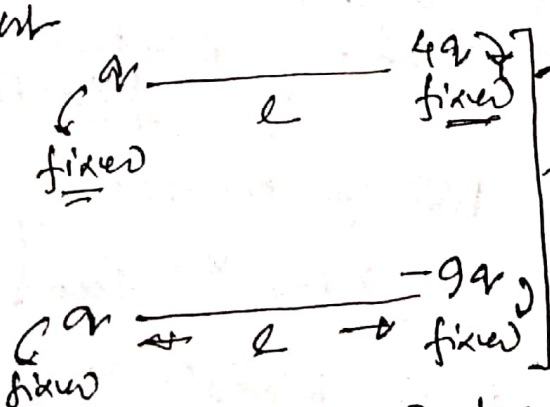
(11)



Find m if two balls are in equilibrium.?

$$\left[\frac{q^2\cot\theta}{16\pi\epsilon_0 g l^2 \sin^2\theta}\right]$$

(12)



Find position of a third charge on the line joining (12) q & 4q so that it can be at equilibrium. (l/3 from q (12)) ($\frac{l}{2}$ on left side of q (13))

(13)

(14) Two very small pendulum bobs (each of density 1.6 gr/cc) having same mass & +ve charge are suspended from a pt in air. If on immersing them in a liquid of density 0.8 gr/cc, Angle betn threads remains same, then find dielectric constant 'K' of liquid (2)

Scanned with CamScanner